

Brain Serotonin Synthesis and Na^+, K^+ -ATPase Activity are Increased Postnatally after Prenatal Administration of L-Tryptophan*

JORGE HERNÁNDEZ-RODRÍGUEZ¹ and GRACIELA CHAGOYA²

Centro de Investigación y Estudios Avanzados del IPN, Laboratorio de Neurontogenia, México, D.F. 07000 (Mexico)

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The effect of prenatal L-tryptophan supplementation on the serotonin (5-HT) synthesis and the activity of Na^+, K^+ -ATPase in the cerebral cortex was studied during postnatal development, from birth up to day 30. A parallel and significant elevation of the serotonin content and the activity of tryptophan-5-hydroxylase was observed in the brain of infant rats born to mothers treated with L-tryptophan, as related to non-treated controls. The activity of Na^+, K^+ -ATPase was also significantly elevated at the different ages studied throughout the developmental period, as related to controls. These results suggest an important role of L-tryptophan in the early regulation of the serotonin-synthesizing machinery, which lasts postnatally. Elevation of ATPase activity seems to be associated to the elevation in the activity of the 5-HT system.

INTRODUCTION

Early malnutrition is known to produce a broad range of non-specific changes in the developing and adult brain^{11,12,22,27}. However, as we and others have shown, its effect on serotonin (5-HT) metabolism in the brain appears to be more specific^{14,15,25,30}. The free fraction of plasma L-tryptophan (L-Trp) has been suggested to cross the blood-brain barrier and stimulate the synthesis of brain serotonin (5-HT)^{13,25,31,32}, in the normal and malnourished rat. An increase in brain 5-HT content in developmentally malnourished rats^{15,30} possibly mediated by an increase only in free plasma Trp²⁵ has been one of the most interesting observations concerning the possible role of essential aminoacids in brain neurochemical development. Nevertheless, other studies have shown that different factors may be involved in the regulation of brain Trp uptake under different physiological conditions. For example, Madras et al.²¹ reported a dissociation between free L-Trp in blood and Trp levels in brain in rats, after glucose loading. The affinity of

Trp for binding to albumin seems to be too low to compete with the Trp transport mechanism²⁶. Fernstrom et al.⁸ postulated that Trp must also compete with other neutral aminoacids (phenylalanine, leucine, isoleucine, valine, etc.) to be transported across the blood-brain barrier by a neutral aminoacid carrier. So, it seems that in various ways plasma Trp would influence brain Trp levels and 5-HT synthesis. During development of a malnourished rat, the mechanisms mentioned above could be also involved in the increase of brain 5-HT observed.

On the other hand, we observed that intraperitoneal (i.p.) administration of 5-HT precursors like Trp and 5-HT induces a response on the Na^+, K^+ -ATPase activity in the rat brain¹⁶. The activity of this important enzyme system is also elevated in the brain of early malnourished rats^{16,17}. Thus changes in 5-HT and ATPase activity appear to be specific alterations resulting from early protein caloric malnutrition, where an important step seems to be an increase in the free plasma Trp, together with a possible change in the other mechanisms involved. Furthermore, on

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Correspondence: J. Hernández, Centro de Investigación y Estudios Avanzados del IPN, Laboratorio de Neurontogenia, Apartado Postal 14-740, México, D.F. 07000, México.

the basis of other results^{16,18}, we have postulated that these two systems possibly interact in the brain.

In the present work we were interested in testing whether early changes in Trp availability could induce similar alterations on 5-HT metabolism and Na^+, K^+ -ATPase activity during postnatal brain development. Thus the effect of L-Trp supplementation on 5-HT synthesis and Na^+, K^+ -ATPase activity during the gestational period was studied.

MATERIALS AND METHODS

Female nulliparous albino rats (Wistar strain) from our local animal facilities were adapted during two weeks to standard environmental conditions: light from 07.00 to 19.00 h; temperature $22 \pm 2^\circ \text{C}$; relative humidity 50–60%. The rats received a balanced pellet diet (Purina) and water ad libitum. After mating females were kept throughout gestation under the conditions described. A supplement of 50 mg/kg body weight of Trp (Sigma, diluted in acidic solution, pH adjusted to 6.8 for injection) was administered i.p., twice a week, during the gestational period. The injection volume administered to each rat did not exceed 0.3% of body weight. The treatment of animals was always between 09.00 h and 11.00 h. Control females were administered an equivalent volume of saline the same days and others were not injected at all (basal values). In the offspring of these groups Na^+, K^+ -ATPase activity, 5-HT content and tryptophan-5-hydroxylase (T5-H) activity were determined in the cerebral cortex, from day zero (day of birth) up to day 30. Pups remained with the mother up to day 30³.

The activity of Na^+, K^+ -ATPase was determined in individual cerebral cortices in all cases, except on day zero in which 4 cortices were pooled. Animals were sacrificed always between 09.00 h and 12.00 h by neck fracture, the brain was immediately perfused with cold saline and the cerebral cortex dissected out at $0-4^\circ \text{C}$. The assay used to measure the enzyme activity is described elsewhere¹⁷. Briefly: about 700 μg of protein homogenate was added to the following components of the incubation mixture: MgCl_2 , 3 mM; KCl, 7 mM; NaCl, 100 mM; ouabain, 0.6 mM; Tris-HCl, 50 mM, pH 7.4. This mixture was preincubated 15 min at 37°C Tris-ATP 4 mM was then added and incubation proceeded for 30 min in a

Dubnoff shaker. Reaction was stopped by adding 1 μl of cold 10% trichloroacetic acid. After centrifugation P1 was determined colorimetrically in the supernatant. The difference between ouabain blanks and total activity was considered to be Na^+, K^+ -ATPase. Proteins were determined by the method of Lowry¹⁹ using bovine serum albumin as the standard. Serotonin concentration in the cerebral cortex was determined, after its separation in a cation exchange resin by the orthophthaldialdehyde (OPT) and cysteine reaction^{6,24}. Briefly: brain tissue (0 days, 4 cortices; 5 days, 3; 10 days, 2) was homogenized in 5 vols. perchloric acid 0.4 N, in the presence of EDTA and ascorbic acid. The homogenate was centrifuged and the supernatant adjusted to pH 6.1 and diluted, after stabilization it was centrifuged again and the supernatant was passed through a cation exchange resin (Bio Rex-70, 200–400 mesh, Na^+ form prepared pH 6.1). Columns were washed with phosphate buffer 0.2 M, pH 6.1, and deionized water. Serotonin was then eluted with normal acetic acid. Aliquots of the eluant were submitted to the OPT reaction to obtain a fluorescent serotonin derivative⁶ which was quantified in an Aminco-Bowman spectrophotofluorometer at 360 and 470 nm activation and emission wavelengths, respectively. A standard curve known 5-HT concentrations was run together with appropriate internal standards and blanks. Recovery in our hands averaged 60%.

For the determination of T5-H activity in the cerebral cortex, the method of Gal and Patterson¹⁰ was used. Tissue (0 and 5 days, total pool of 5 cortices) was homogenized 3:1 (v/w) in Tris acetate, 0.5 M, pH 7.6, containing 2-mercaptoethanol 10^{-3} M. The incubation mixture contained: Tris acetate buffer 5 mM, pH 7.6; 2-mercaptoethanol 1 mM; pargyline 0.2 mM; L-Trp 0.088 mM; 6-methoxytetrahydrobiopterin 0.16 mM; 10 mg catalase and 0.5–3 mg protein homogenate. The mixture was incubated 5 min in a metabolic shaker at 37°C and the reaction was stopped by heating samples 5 min in a boiling waterbath. Then the samples were allowed to react with cysteine (1%) and OPT in the presence of 10% HCl. The 5-hydroxyindols formed were measured in an Aminco-Bowman spectrophotofluorometer at 360 nm and 470 nm, activation and emission wavelengths, respectively. Appropriate standards and blanks were included in the assay.

RESULTS

There was a developmental increase of 5-HT content in the cerebral cortex of normal controls, from birth up to day 21, stabilizing thereafter to adult values. In the cerebral cortices of infant rats born from mothers treated with Trp there was a significant increase in the content of 5-HT ranging from 37 to 13% on days 0, 5, 10, 15, 21 and 30 as related to saline controls (Fig. 1). The saline controls showed non-significant intermediate values. The activity of T5-H, the rate-limiting enzyme in the synthesis of 5-HT, was already present at birth (about 10 percent of the adult values) and increased progressively up to day 30 in the normal controls. T5-H activity was significantly elevated in rats from mothers treated with Trp from day 5 up to day 30, as compared to normal and saline controls, (Fig. 2). Acute Trp administration to gestational rats also induced a response of T5-H activity in the cerebral cortex of mothers and fetuses; (see Table I).

Na⁺,K⁺-ATPase activity showed a slight, but not significant increase over control values from birth up to day 10. However, from day 15 up to day 30 a significant increase was detected, in relation to saline and basal controls (Fig. 3).

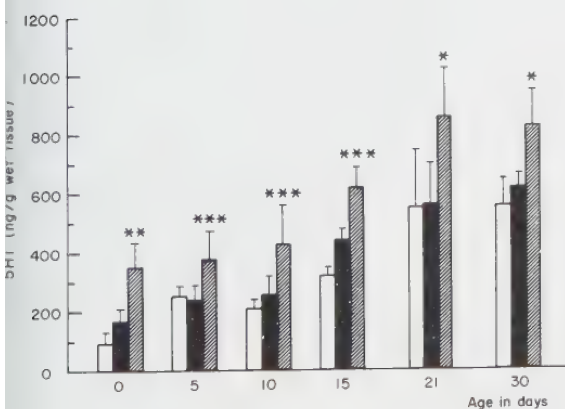


Fig. 1. Serotonin content in the cerebral cortex of rats born from mothers treated during gestation with L-Trp 50 mg/kg, i.p., 6 ×. □, Basal controls (animals minimally handled, mothers not injected); ■, vehicle-injected controls; ▨, L-Trp-treated animals. Each bar represents mean values from 3 to 11 different experiments ± S.D. For each experiment in young animals several cortices were pooled (see Materials and Methods) and pups were always taken from at least 3 different litters. Determinations were made in duplicate samples (* $P < 0.05$; ** $P < 0.01$, *** $P < 0.001$; Student's *t*-test as related to controls).

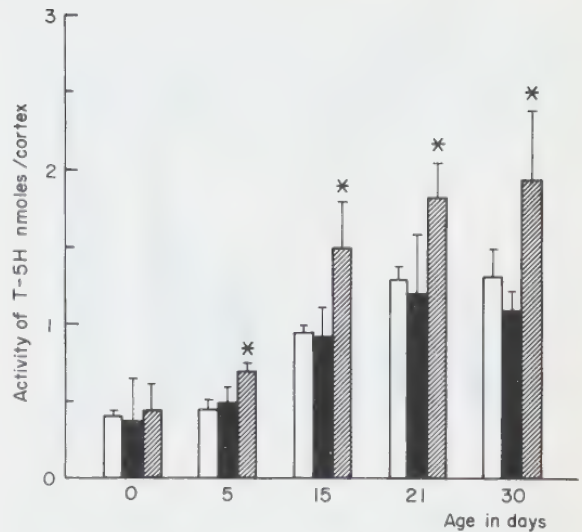


Fig. 2. Postnatal development of T5-H activity in the cerebral cortex of rats born from mothers treated during gestation with L-Trp 50 mg/kg, i.p., 6 ×. □, Basal controls (animals minimally handled, mothers not injected); ■, vehicle-injected controls; ▨, L-Trp-treated. Bars represent mean values from 3 different experiments ± S.D. For each experiment in young animals several cortices were pooled (see Materials and Methods) and pups were always taken from at least 3 different litters. Determinations of enzyme activity were made in triplicate samples. (* $P < 0.05$; Student's *t*-test as related to controls).

Other results (not shown), in which 5-hydroxyindoleacetic acid (5-HIAA) levels in the brain were measured in animals treated similarly with L-Trp during gestation, also suggest that this 5-HT metabolite is increased in the Trp-treated group.

DISCUSSION

The developmental pattern of the 3 parameters studied here showed an intensive increase during the

TABLE I

Activity of T5-H

T5-H activity in the brains of pregnant rats and fetuses 30 min after the acute administration of L-Trp, 100 mg/kg, i.p., and saline controls, on day 15 of gestation. Activity is expressed as nmol indoles/mg protein/h. Numbers are mean values from 3 determinations ± S.D. (for fetuses a pool of about 30 brains was made, enzyme activity was measured in triplicate samples). * $P < 0.05$; ** $P < 0.01$; Student's *t*-test.

Brain	L-Trp	Saline
Mother	0.354 ± 0.024	0.253 ± 0.039**
Fetus	0.257 ± 0.038	0.182 ± 0.028*

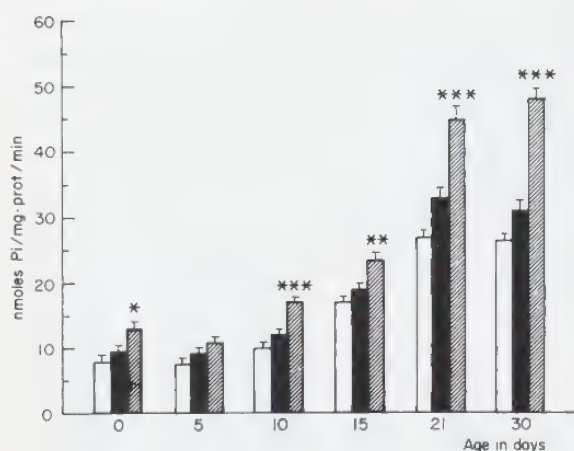


Fig. 3. Postnatal development of Na^+, K^+ -ATPase activity in the cerebral cortex of rats born from mothers treated during gestation with L-Trp 50 mg/kg, i.p., 6 $\times$. □, Basal controls (animals minimally handled, mothers not injected); ■, vehicle-injected controls; ▨, L-Trp-treated animals. Bars represent mean values from 3 to 11 different experiments $\pm$ S.D. For each experiment in young animals several cortices were pooled (see Materials and Methods) and pups were always taken from at least 3 different litters. Determinations of enzyme activity were made in triplicate samples. (* $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; Student's *t*-test as related to controls).

second and third week of postnatal life. This period corresponds to the maturation of 5-HT nerve terminals¹⁹, the formation of specific 5-HT receptors⁹, the maturation of electrical activity⁵ and synaptogenesis² in the rat brain. A further increase in 5-HT content in the brain of older rats has been reported by Bennett and Giarman⁴, in the present work the period studied was only up to postnatal day 30. Deguchi and Barchas⁷ observed that maturation of T5-H activity occurred in the rat cerebral cortex somewhat later, about day 30. However, Schmidt and Sanders-Bush²⁸ described an important increase of T5-H activity during the second and third postnatal week, but they report a decrease in the adult. Concerning Na^+, K^+ -ATPase activity, Meisami and Timiras²³ reported brain regional increases between day 10 and 25 in the inferior colliculi, and up to day 40 in frontal cortex. Abdel-Latif et al.¹ observed an intensive increase of enzyme activity in nerve-ending fractions up to day 15 of postnatal life. Strain and methodological differences could account for the comparative discrepancies.

The present results indicate that the availability of an essential aminoacid, L-Trp, during the gestational

period seems to be of relevance to brain neurochemical development. This precursor molecule appears to regulate the biosynthesis of its final functional derivative, the neurotransmitter 5-HT, in the developing brain. This conclusion is supported by a parallel increase in the brain 5-HT content and the activity of the rate-limiting enzyme T5-H in the brain of developing rats treated with Trp during gestation. The acute administration of Trp to pregnant females produced an increase in T5-H activity in the brains of both the mother and the embryo. It is not clear at the moment if this early stimulus on differentiating 5-HT cells accounts for the response observed postnatally on 5-HT synthesis. Possibly a developmental increase in 5-HT release and turnover is associated since the 5-HIAA content appears to be also elevated, suggesting a possible increase in the serotonergic neuronal activity whose functional relevance remains to be studied. However, an effect of the early Trp administration on other neurotransmitter systems cannot be ruled out. Smith et al.²⁹, reported an effect of acute Trp administration, not only on the amount of brain 5-HT, but also on the aminoacids aspartate, glutamate and glycine.

It is difficult to explain at the moment the mechanisms underlying the postnatal response observed; it could be possible that changes on the 5-HT biosynthetic machinery are provoked by the early Trp administration. It is known that in the adult brain T5-H has a low affinity for Trp resulting in a limitation of 5-HT synthesis. Early Trp administration could account for a shift to a high affinity of the enzyme for its substrate, resulting in a higher rate of 5-HT synthesis thereafter. A study of T5-H kinetic constants before and after treatment with Trp could provide information about this point. Zeisel et al.³³, have shown that fasting, in young rats, induces higher concentrations of serum Trp and higher serum Trp/large neutral aminoacids ratios; this, together with an increase in brain Trp and 5-HIAA, suggests a regulatory role of dietary Trp on brain 5-HT synthesis during the developmental period.

The developmental changes in the 5-HT system are associated to the activation of another important system, the Na^+, K^+ -ATPase. An explanation of this effect can be proposed in the light of previous results showing a possible interaction between the two systems. Indeed, results reported from our laboratory

y^{17,18} show that stimulation of the brain 5-HT system with agonists such as Quipazine induces a significant increase of brain Na⁺,K⁺-ATPase activity in adult and developing rats. Furthermore, in the brain of developmentally malnourished rats we have observed both an increase in 5-HT content and an elevation of

Na⁺,K⁺-ATPase activity¹⁶. Therefore a close association between brain 5-HT and Na⁺,K⁺-ATPase changes could be a consequence of the early stimulation on the serotonergic system by Trp administration.

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